

Technical Module Mapping (React + Node.js LMS)

Module	React (Frontend)	Node.js (Backend)
Authentication & Authorization	Login/Register Pages, Route Guards, Role-based Navigation	JWT Middleware, Login APIs, Role Validation
User Management	Student/Teacher/Admin Dashboards, User Lists	User CRUD APIs, Role Assignment, Approval Logic
Course Management	Course Listing, Create/Edit Course Pages	Course CRUD APIs, Course–Teacher Mapping
Lesson Management	Lesson List UI, Video Player, Notes Viewer	Lesson CRUD APIs, Video & Notes Linking
Content Delivery	Video Streaming UI, PDF Viewer	Content Fetch APIs, Media Access Control
Quiz & Assessment	Quiz UI, Timer, Result Page	Quiz Creation APIs, Auto-Evaluation Logic
Attendance Management	Attendance Marking UI, Attendance Reports	Attendance APIs, Date & Status Tracking
Assignment & Homework	Assignment Upload UI, Submission Page	File Upload APIs, Submission & Grading Logic
Announcement & Notice Board	Notice Feed UI, Notification Panel	Announcement APIs, Broadcast Logic
Progress Tracking & Analytics	Charts, Progress Dashboards	Analytics APIs, Performance Calculation
Gamification	Badges UI, Leaderboard Page	Reward Logic APIs, Score Calculation
Certificate Management	Certificate Preview & Download	Certificate Generation APIs
Discussion / Doubt Forum	Discussion Threads UI, Comment Box	Forum APIs, Moderation Logic
Admin Control Panel	Admin Dashboard, System Settings UI	Admin APIs, Platform Configuration
File & Media Management	File Upload Components, Media Preview	File Storage APIs, Access Control
System Settings	Settings UI, Language Selector	Configuration APIs
Security & Audit	Activity Logs UI	Audit Trail APIs, Access Logs